

ME 300 Homework 1: Energy Conservation and Basic Principles

Answer the following questions neatly on a separate piece of paper or on a tablet. Written answers can be typed. Show all your work and explain your reasoning – partial credit will only be given if you show and explain each of your steps. Submit your homework on Canvas.

1. (10 points) Please fill out the energy survey on Canvas. This is a *survey* to get a baseline of your understanding of energy concepts at the beginning of the class. **Do not look up any of the answers online!!!** You will get full credit for simply answering the questions, so just do your best.
2. (10 points) Again, without looking up the answers, explain how electricity is produced and gets to your appliances. Let's use your coffee pot as an example – work back from the plug of your coffee pot all the way to the original source of energy. Where does electricity come from and does it get to your coffee pot? Answer this question to the best of your ability without any resources, it will be graded on completion only.
3. (30 points) Consider these five conservation scenarios and provide the following information about how to solve for the scenario:
 - Indicate whether it's appropriate to use a finite-time or instantaneous conservation formulation.
 - Write the form of the conservation equation, cancelling any terms that are not required and explaining why these terms can be neglected.
 - a. One way to conserve water is to catch it in a rain barrel to water your plants and your lawn (<https://extension.psu.edu/rain-barrels-for-water-conservation>). In this scenario, a rain bucket with a volume V collects rain at a rate of $\dot{m}_{rain}$ during a rain storm. Assume that the bucket doesn't overflow. Write a conservation of mass statement for the water in this scenario.
 - b. You want to make coffee using a Mr. Coffee (classic), which heats the water in the well at a rate of $\dot{Q}_{heat}$, but the coffee machine loses heat at a rate of $\dot{Q}_{loss}$. Write a conservation of energy statement for the water in this scenario.
 - c. Just as you're about to make more coffee, you find coffee still in the pot from the morning – bonus! You put it in a mug and put it in the microwave for 1 minute, where Q_{in} heat is added to the cup. Write a conservation of energy statement for the coffee in this scenario.
 - d. Natural gas is used in a wide range of applications, from power generation to home heating. In order to move natural gas around the country, there is a network of gas pipelines where gas flows from wells to local distribution networks. In order to keep the gas moving, you need to “recompress” the gas along the pipeline

in a gas recompression station: <https://extension.psu.edu/understanding-natural-gas-compressor-stations> Here, gas flows through at a rate of $\dot{m}_{gas}$ and a compressor does work on the gas to raise its pressure at a rate of $\dot{W}_{compressor}$. Write a conservation of energy statement for the gas in this scenario.

- e. Penn State has a solar farm on campus that can produce up to 2 MW of power – learn more here: <https://www.psu.edu/news/campus-life/story/project-provide-reliable-and-affordable-solar-generated-energy-penn-state-0/> Although our solar farm doesn't have one, many new solar installations have battery packs to store the energy. In this system, E_{solar} Joules of energy would be captured by the solar panels each day and Q_{loss} Joules of energy would be lost from the system through electrical inefficiencies during the same time period. During the day, the power goes out and the house uses E_{house} Joules of energy. Write a conservation of energy statement for the energy in the battery during the day.
4. (30 points) Consider the five scenarios from the previous question and answer the following:
 - Indicate whether we should use a system or control volume analysis and why.
 - Draw the scenario and then draw the system or control volume
 - Label the system/control volume, boundary/control surface, and surroundings
 - Label the energy and/or mass flows in and/or out of the system/control volume
 5. (10 points) Identify the proper SI unit for each property – use the book as a reference:
 - a. Pressure
 - b. Temperature
 - c. Volume
 - d. Energy
 - e. Power
 - f. Length
 - g. Time
 - h. Mass
 - i. Velocity
 - j. Density
 6. (10 points) I like to get to know the students in my class each semester! Please tell me why you've chosen mechanical engineering as your major (or whatever your major is), and what you'd like to do in your career after college. What types of technologies are you interested in and might like to hear more about in this class? I'll read every one of these responses!